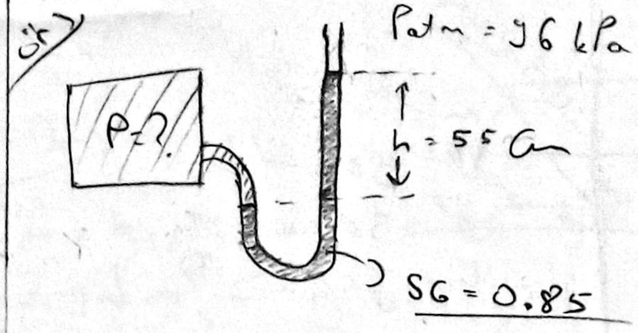
Termo
Problem C

①

Ör)


P=?
Patm = 96 kPa
h = 55 cm
SG = 0.85

$P = P_{atm} + \rho . g\, h$

$P = 96\,kPa + 0{,}85 \times 1000 \times 9{,}81 \times \bigcirc\!\!\!\!\!\!\; 0{,}55\,m$ ✓

metreye çevir.

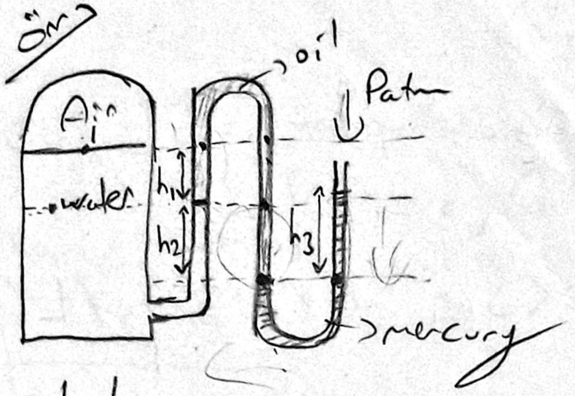
$P = 100.6\,kPa$

$P_{air} + P_{water} + P_{oil} \pm P_{mercury} + P_{atm}$

$P_{air} = \rho_{su} \cdot g \cdot h_2 + \rho_{oil} \cdot h_1 \cdot g = \rho_{mercury} \cdot g.$

$P_{air} = P_1$

Ör)


Air
water
oil
Patm
h1
h2
h3
mercury

verilenler

$P_{atm} = 85{,}6\,kPa$

$h_1 = 0{,}1\,m \quad h_2 = 0{,}2\,m \quad h_3 = 0{,}35\,m$

$\rho_{su} = 1000\,kg/m^3,\ \rho_{yağ} = 850\,kg/m^3,\ \rho_{cıva} = 13600\,kg/m^3$

$P_{air} = ?$

$P_{air} + P_{water} + P_{oil} h_2 = P_{atm} + P_{mercury}$

$P_{air} + \rho_w g\, h_1 + \rho_{oil}\, g \cdot h_2 = P_{atm} + \rho_m g \cdot h_3$

~~85600 Pa~~

$P_1 = 85{,}6\,kPa + 13600 \cdot 9{,}81 \cdot 0{,}35\,m - 1000 \cdot 9{,}81 \cdot 0{,}1 - 850 \cdot 9{,}81 \cdot 0{,}2\,m$

$P_1 = 129{,}6\,kPa$

Örn 2-1

Günlük harc. benzin = 5L/Gün

Depo = 50L

$0{,}68 < \rho_{Benzin} < 0{,}78$ kg/L

$Q_{yanma} = 44000$ kJ/kg

U-235

0,1kg U-235

=10 gün yeter mi.

$44000 \, kJ/\cancel{kg} \cdot 0{,}78 \, \cancel{kg}/\cancel{L} \cdot 50\cancel{L}$

$Q_{toplam} = 1.716.000$ kJ

$Q_{günlük} = 171.600$ kJ

Örn 2-2

$\dot{m} = 8{,}5$ m/s

~~K~~E = ?

$$\cancel{m} \cdot \frac{V^2}{2} = \frac{(8{,}5)^2}{2} = \overset{ke}{36{,}12} \, \cancel{m^2/s^2} \, J/kg$$

Bi'e bir dönüşüyor.

b) m = 10 kg

$36{,}12 \times 10 = 361{,}2 \, J$

c) 1154 kg/s ⇒ $36{,}12 \times 1154$ kg/s = [J/s]

kW

Örn 2-8

m = 1200

~~V~~ = 90 km/h

$\alpha = 30^\circ$

ilave güç → ? hız sabit kalması için

$$\Delta W = \frac{m \cdot g \cdot \Delta Z}{\Delta t} = \rightarrow V_{dikey}$$

90 km/h
30°
?
Cevap

$\Delta W = 1200 \, kg \times 9{,}81 \, m/s^2 \times \sin 30 \cdot 90 \, km/h$

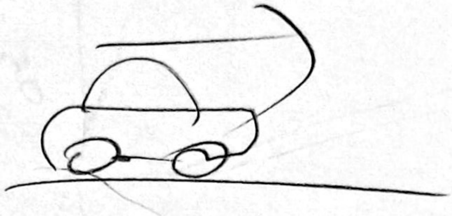
$1200 \, kg \times 9{,}81 \, m/s^2 \times \sin 30 \times 90 \, \cancel{km}/\cancel{h} \times \frac{1000 \, m}{1 \, \cancel{km}} \times \frac{1 \, \cancel{h}}{3600 \, s}$

$\Delta W = 147150 \, [m^2/s^2 = J/s]$

$= 147{,}150 \, [kJ/s = kW]$

Team Problem ②

Örnek 2-9



m = 900kg

t = 20 s.

$\int_{0}^{80\,km/h} V = ?$

$\frac{0-80\,km/h}{20s} = a$

$\Delta W = \frac{1}{2} \cdot m \left(\overset{80}{V_2} - \cancel{\overset{0}{V_1}}\right)^2$

$W = \frac{1}{2} \cdot 900kg \times (80\,km/h)^2 \times \frac{10^6 m^2}{1 km^2} \times \frac{1h^2}{(3600)^2 s^2} \left(\times \frac{1000 J}{1 kJ}\right)^{-1} \times \cancel{\left(\frac{1 J/kg}{1 m^2/s^2}\right)}$

W = 222,2 kJ

$\text{Work} = \frac{222,2\,kJ}{20\,sn} = 11,11 \boxed{kJ/s = kW}$

Örn 2-13

80W × 10 floresan

$Q_T = 2400\ W$

250g × 12h = 3000h

1 kW-h elektrik → 7 sent

2,4 kW × 3000h = 7200 kW-h × 7

$50400\ \cancel{sent} \times \frac{1\$}{100\ \cancel{sent}} = 504\ \$/\text{yıllık}$

✓

Örn 2-15

Ocak elektrikli → Gaz

η = 73 %   η = 38 %

birim/elektrik = \$0,09/kW-h   \$0,55/therm   [illegible]

2kW enerjili bir Ocak için maliyetler.

~~$2\ kW \times \frac{1}{73} \times 0{,}09/kW\text{-}h$~~ (m)

= 0

$Camp\left[\frac{0{,}09\ \$}{\eta} = Camp.\right.$

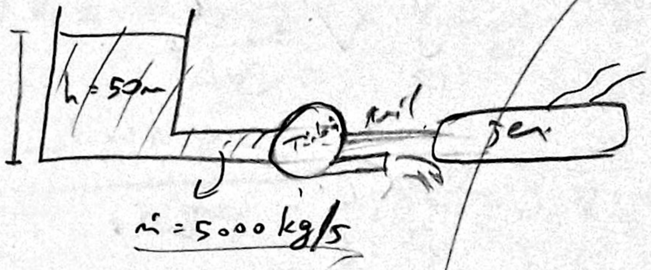
Örn 2-16

a) η_{Topla}
b) $\eta_{Türb.}$
c) mil gücü

$\dot{m} = 5000\ kg/s$

$\dot{W}_{elektrik} = 1862\ kW$

$\eta_{jen} = \%95$



c) $\frac{1862\ kW \times 100}{95} = 1960\ kW$

b) $\dot{m} \cdot g \cdot \Delta z$ 9,81 × 50 m nete

$5000\ kg/s \times 9{,}81\ m/s^2 \times 50\ m \times \frac{1\ J/kg}{1\ m^2/s^2} \times \frac{1\ kJ}{1000\ J}$

2452,5 kg·kJ/s = kW

$\frac{1960\ kW}{2452{,}5\ kW} = 0{,}8 \cdot \eta_{Türb.}$ ✓

$\eta_{Topl} \Rightarrow$

a) $Q_a / Q_g \Rightarrow \left(\frac{2452{,}5}{1862}\right)^{-1} = 0{,}76$

Termo Problemler ③

Soru 1) Const. Valve

$T = 90°C$

$m = 50\ kg$

doymuş sıvı hal su

$P = ?$ $V = ?$

@ A-~~5~~ ④. → P ve V değerleri

$P = \cancel{75}\ kPa$
70

$v_f = 0{,}001036\ m^3/kg \times 50\ kg$

$V = 0{,}0518\ m^3$

Soru 2)

$m = 200\ g = 0{,}2\ kg$

$P = 100\ kPa$

doymuş sıvı su buharlaştırılıyor.

a) Hacim değişikliği

b) Sağlanan enerji.

@ Tablo A-5 ~~T ve~~ v_f ve v_g değerleri.

~~T=~~ $v_f = 0{,}001043$ $v_g = 1{,}6941\ [m^3/kg]$

Hal değişim enerjisi ~~=> $\Delta U = U_g - U_f$~~ $u_g = 2505{,}6\ [kJ/kg]$

$u_f = 417{,}40$

a) $V_f = v_f \times m = 0{,}2\ kg \times v_f$

$V_f = 2{,}086 \times 10^{-4}\ m^3$

$V_g = 1{,}6941\ m^3/kg \times 0{,}2\ kg$

$V_g \cong 0{,}339\ m^3$

$V_g - V_f = \Delta V \cong 0{,}3386\ m^3$

~~b) $\Delta U_g = U_g - U_f = (u_g - u_f)(m)$~~

~~$\Delta U = 0{,}2 \times (2505{,}6 - 417{,}40)\ kg \times \frac{kJ}{kg}$~~

~~$\Delta U = 417{,}64\ kJ$~~

Δh hesaplanacaktır.

↳ Hacim sabit olmadığından ekstra enerji gerekiyor.

$\Delta h = \Delta U + \boxed{P \Delta V}$ → 100 kPa × 0,338

bunu

örn

Const. Volume

T = 90°C   a) = P = ?

m = 10 kg su   b) = V = ?

8kg sıvı 2kg buhar

@ T = 90°C için Tablo A-4

a) $P_{doyma} = 70{,}183$ kPa $v_f = 0{,}001036$

$v_g = 2{,}3593$

b) $V_T = v_f \times 8 + v_g \times 2$

$V_T = 0{,}001036 \times 8\ \cancel{kg} \times \cancel{m^3/kg} + 2{,}3593 \times 2$

$V_T = 4{,}727\ m^3$

~~x~~ = yüzde buhar kütlesi ⟹

$m_T \cdot x = \cancel{m_g}$ $m_g = 0{,}628$ kg

$V_g = 0{,}628\ \cancel{kg} \times v_g\ m^3/kg$ $v_g = 0{,}12348$

$V_g = 0{,}0775\ m^3$

örn

$V_k = 80\ Lt$ = kap.

$P = 160\ kPa$

m = 4kg soğutucu Akışkan-134a

a) $T_a = ?$ b) $x = ?$ c) $h = ?$ D) $V_{buhar} = ?$

$80\ Lt \times \frac{1 m^3}{1000\ Lt} = 0{,}08\ m^3 = V_{kap.}$

m = 4 kg

$v = 0{,}02$ ve $P = 160\ kPa$ @ A-12

$T = -15{,}60$ ⓐ) $v = v_f + v_{fg} \cdot x$

$v_f = 0{,}0007437\ [m^3/kg]$

$0{,}02 = 0{,}0007437 + (V_g - V_f)x$

$v_g = 0{,}12348$

$0{,}02 = 0{,}0007437 + 0{,}1227363\, x$

$h_f = 31{,}21$

$h_{fg} = 209{,}90\ [kJ/kg]$

$x = \frac{0{,}02 - 0{,}0007437}{0{,}1227363}$

x = kütlelerin oranı

b) $x = 0{,}15689$

$h = h_f + h_{fg} \cdot x$

$h = 64{,}141\ [kJ/kg]$ ⓒ

D) ~~x~~ $= \frac{V_b}{V_T}$ $\Rightarrow V_b \neq x \cdot V$

hacimlerin oranı

$V_b = 0{,}08 \times 0{,}15689$

$V_b = 0{,}01255\ m^3$

Termo I
Problem ⓐ

Ör) $P = 1\ MPa$
$T = 50°C$
Soğutucu Akışkan-134a
$v = ?$ $v_{gerçek} = 0{,}02179$ ile kıyasla

a) $Pv = RT$ ile   b) Genelleştirilmiş sıkıştırılabilirlik diyagramı

ⓐ Tablo A-1 ~~R-12~~

$R = 0{,}06876\ kJ/kg \cdot K°$
$\cong 0{,}0815$

$10\ kPa \times V = 0{,}06876\ kJ/kg \cdot K° \times 323\ K°$

$kPa = N/m^2$
$kJ = N \cdot m$

$v = \frac{0{,}06876 \times 323}{10}\ m^3/kg$

$2{,}22\ m^3 \times 10^{-3}$ Dikkat

$1\ MPa = 1000\ kPa$

$v = \frac{0{,}0815 \times 323}{1000} = 0{,}0263\ m^3$

$\frac{0{,}0263 - 0{,}0217}{0{,}0217} = 0{,}213$ hata Oranı

devam

$P_{cr} = 4{,}059\ MPa$
$T_{cr} = 374{,}2\ K$

$P_R = \frac{P}{P_{cr}}$, $T_R = \frac{T}{T_{cr}}$

$\frac{1\ MPa}{4{,}059\ MPa} = 0{,}246$
$\frac{323\ K}{374{,}2\ K} = 0{,}86$
$\Big\} Z = 0{,}84$

@ Tablo A-15
$P_R = 0{,}246$
$T_R = 0{,}86$ için

$v = v_{ideal} \times Z$
$v = 0{,}0220$

$\frac{0{,}0220 - 0{,}0217}{0{,}0217} = 0{,}014$
%1,4

$x = 60$

300° [illegible]

$h = ?$

1,6 MPa — 3035,4.
1,7 MPa — x
1,8 MPa — 3029,9

$$\frac{x - 3035,4}{-(3035,4 - 3029,9)} = \frac{1,7 - 1,6}{1,8 - 1,6} = \frac{0,1}{0,2}$$

$x = 3032,65 \; [kJ/kg]$

x değeri (kuru buhar oranı) = %100
[illegible]